

Question

X-ray diffraction reveals that the anion of the diamagnetic pink potassium salt $\text{K}_2\text{Mo}(\text{SO}_4)_2$ is a binuclear complex ion, in which the sulfate ion acts as a bridging ligand, and the nuclear distance between molybdenum atoms is 211 pm. Recrystallization of this double salt yields a potassium salt with the chemical formula $\text{K}_3\text{Mo}_2(\text{SO}_4)_4 \cdot 3.5\text{H}_2\text{O}$, whose anion retains the structure of the $\text{K}_2\text{Mo}(\text{SO}_4)_2$ anion but exhibits a different overall charge. Calculate the product of the Mo – Mo bond orders for the two anions mentioned above.

A. 0.5

B. 1.5

C. 3

D. 5

E. 7.5

F. 10.5

G. 14

H. 18

I. 22.5

J. 33

K. 39

L. 12

M. 16

Answer

Correct Answer: **G**

Detailed Explanation

According to the problem statement, the dinuclear anion in $\text{K}_2\text{Mo}(\text{SO}_4)_2$ is $[\text{Mo}_2(\text{SO}_4)_4]^{4-}$, which contains 2 Mo atoms and 4 sulfate groups, with a total charge of -4.

CHECKPOINT

1 PTS

The dinuclear anion is $[\text{Mo}_2(\text{SO}_4)_4]^{4-}$

In this anion, the oxidation state of Mo is $\frac{1}{2}(-4 + 2 \times 4) = +2$, and the number of valence electrons is $6 - 2 = 4$.

CHECKPOINT

1 PTS

The oxidation state of Mo is +2, and the number of valence electrons is 4.

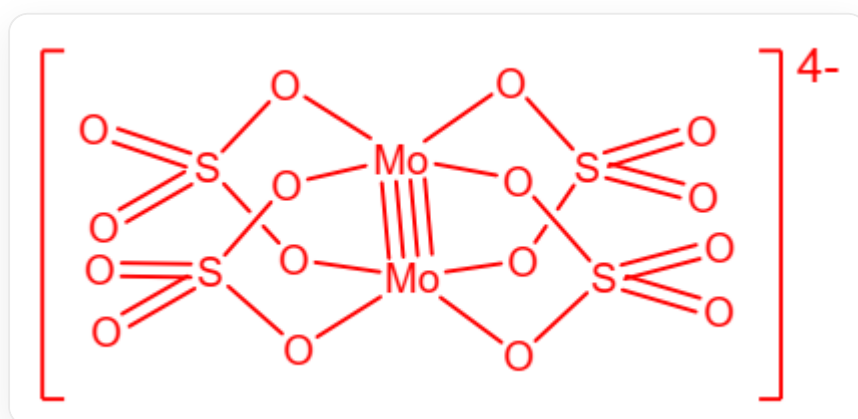
Each Mo atom contributes 4 valence electrons, forming a diamagnetic anion, making it reasonable for a Mo — Mo quadruple bond to form.

CHECKPOINT

1 PTS

The bond order of Mo – Mo in the $[\text{Mo}_2(\text{SO}_4)_4]^{4-}$ anion is 4

Based on the above analysis, the structure of $[\text{Mo}_2(\text{SO}_4)_4]^{4-}$ can be drawn as:



[Structure of $[\text{Mo}_2(\text{SO}_4)_4]^{4-}$, containing 2 Mo, 4 S, 16 O, enclosed by square brackets with a 4- superscript at the upper right. A quadruple bond exists between Mo-Mo, with 4 five-membered rings arranged in the Mo-O-S-O-Mo sequence. The Mo-O and S-O bonds in the rings are single bonds. The 8 O atoms bonded to Mo form parallel and orientationally overlapping squares. Each S is connected to two terminal O atoms via S=O double bonds outside the rings.]

The anion in the crystal $\text{K}_3\text{Mo}_2(\text{SO}_4)_4 \cdot 3.5\text{H}_2\text{O}$ is $[\text{Mo}_2(\text{SO}_4)_4]^{3-}$, which has one fewer electron compared to $[\text{Mo}_2(\text{SO}_4)_4]^{4-}$.

CHECKPOINT

1 PTS

The anion in another salt is $[\text{Mo}_2(\text{SO}_4)_4]^{3-}$

Considering the electronic structure of the central metal in $[\text{Mo}_2(\text{SO}_4)_4]^{4-}$, the 8 coordinating oxygen atoms from the 4 bridging sulfate groups form a square coordination. Each Mo contributes 4 valence electrons, forming a

Mo – Mo quadruple bond, with the corresponding electronic structure being $\sigma^2\pi^4\delta^2$.

CHECKPOINT

1 PTS

The electronic structure of $[\text{Mo}_2(\text{SO}_4)_4]^{4-}$ is $\sigma^2\pi^4\delta^2$.

After losing one electron, $[\text{Mo}_2(\text{SO}_4)_4]^{3-}$ is obtained, with the lost electron originating from the highest-energy δ bonding orbital. The electronic structure becomes $\sigma^2\pi^4\delta^1$, and the bond order is 3.5.

CHECKPOINT

1 PTS

The electronic structure of $[\text{Mo}_2(\text{SO}_4)_4]^{3-}$ is $\sigma^2\pi^4\delta^1$, and the bond order is 3.5.

Thus, the product of the Mo – Mo bond orders for the two anions is $4 \times 3.5 = 14$.

CHECKPOINT

1 PTS

The product of the Mo – Mo bond orders for the two anions is $4 \times 3.5 = 14$.